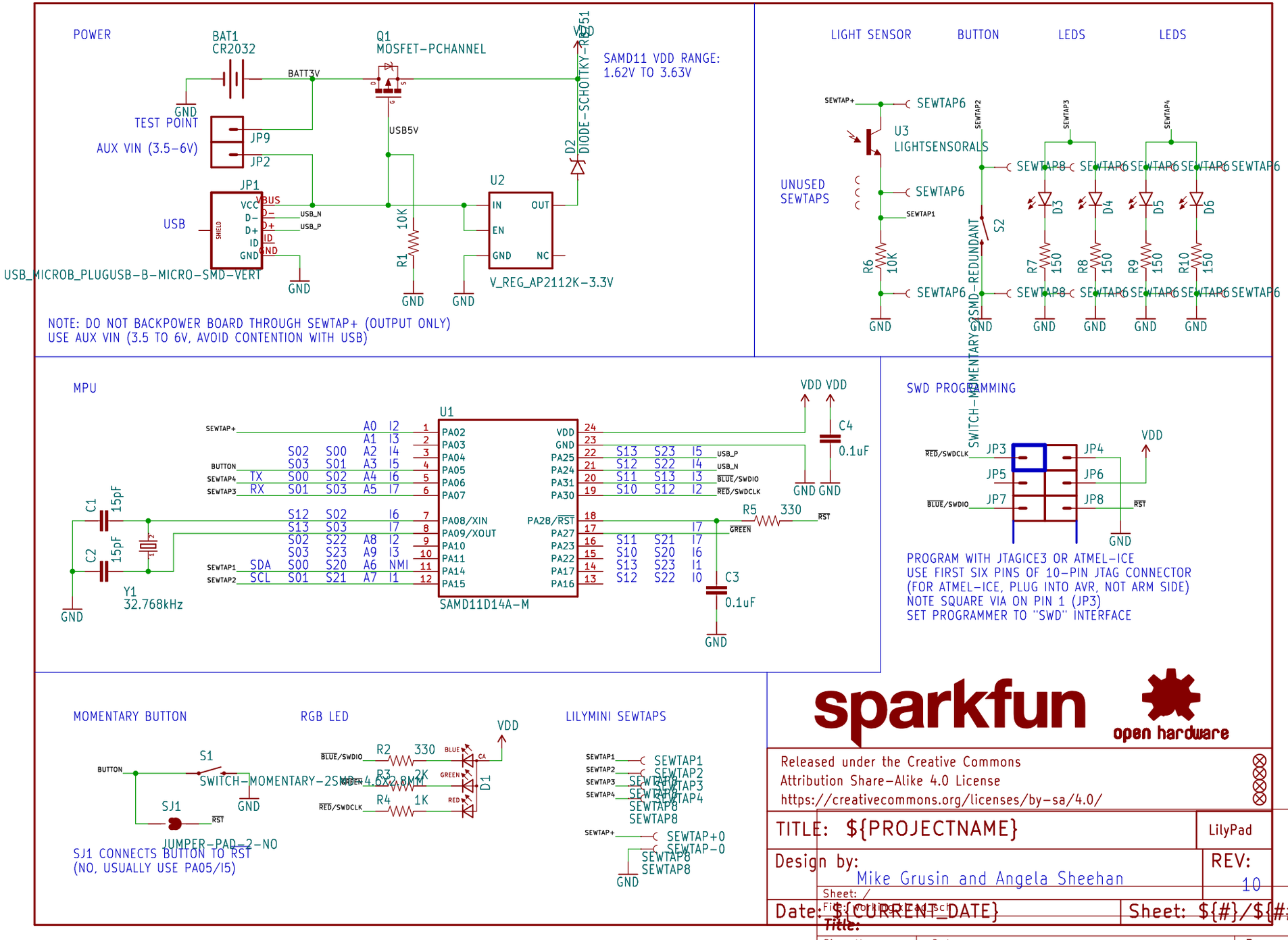
POWER
BAT1
CR2032
BATT3V
Q1
MOSFET-PCHANNEL
VBUS
D2
DIODE-SCHOTTKY-RED
SAMD11 VDD RANGE:
1.62V TO 3.63V
GND
TEST POINT
JP9
AUX VIN (3.5-6V)
JP2
USB5V
JP1
USB
VCC
D-
D+
ID
GND
SHIELD
USB_N
USB_P
USB_MICROB_PLUGUSB-B-MICRO-SMD-VERT
10K
R1
U2
IN
OUT
EN
GND
NC
V_REG_AP2112K-3.3V
NOTE: DO NOT BACKPOWER BOARD THROUGH SEWTAP+ (OUTPUT ONLY)
USE AUX VIN (3.5 TO 6V, AVOID CONTENTION WITH USB)
LIGHT SENSOR
BUTTON
LEDS
LEDS
SEWTAP+
SEWTAP6
U3
LIGHTSENSORALS
SEWTAP2
SEWTAP3
SEWTAP4
SEWTAP8
UNUSED
SEWTAPS
SEWTAP1
S2
SWITCH-MOMENTARY-2SMD-REDUNDANT
R6
10K
D3
D4
D5
D6
R7
150
R8
150
R9
150
R10
150
MPU
U1
SEWTAP+
BUTTON
SEWTAP4
SEWTAP3
SEWTAP1
SEWTAP2
TX
RX
SDA
SCL
PA02
PA03
PA04
PA05
PA06
PA07
PA08/XIN
PA09/XOUT
PA10
PA11
PA14
PA15
VDD
GND
PA25
PA24
PA31
PA30
PA28/RST
PA27
PA23
PA22
PA17
PA16
USB_P
USB_N
BLUE/SWDIO
RED/SWDCLK
GREEN
RST
C1
15pF
C2
15pF
Y1
32.768kHz
C4
0.1uF
C3
0.1uF
R5
330
SAMD11D14A-M
SWD PROGRAMMING
JP3
JP4
JP5
JP6
JP7
JP8
PROGRAM WITH JTAGICE3 OR ATMEL-ICE
USE FIRST SIX PINS OF 10-PIN JTAG CONNECTOR
(FOR ATMEL-ICE, PLUG INTO AVR, NOT ARM SIDE)
NOTE SQUARE VIA ON PIN 1 (JP3)
SET PROGRAMMER TO "SWD" INTERFACE
MOMENTARY BUTTON
RGB LED
LILYMINI SEWTAPS
S1
SWITCH-MOMENTARY-2SMD-4.6X2.8MM
SJ1
JUMPER-PAD_2-NO
SJ1 CONNECTS BUTTON TO RST
(NO, USUALLY USE PA05/I5)
R2
330
R3
2K
R4
1K
D1
BLUE
GREEN
RED
CA
SEWTAP0
SEWTAP-0
sparkfun
open hardware
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TITLE: ${PROJECTNAME}
LilyPad
Design by:
Mike Grusin and Angela Sheehan
REV:
10
Sheet: /
File: Control Board.sch
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
Title:
Size: User
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6~ubuntu22.04.1
Id: 1/1